

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **System Analysis & Design**

Code No: **CACS203** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. You have been hired as a project manager of a large government project to replace an aging air traffic control system. Would you choose waterfall or agile as your application development method? Explain your decision.
3. Explain the roles and responsibilities of software project manager in project development.
4. Giving a simple example, explain how ER schemas are converted into a set of relations. Your example should have the use of multivalued and composite attributes too. Cardinality is a must.
5. What is meant by deliverable? Explain the major deliverables of planning phase.
6. What is the significance of normalization? Explain 2NF giving a suitable example.
7. Explain different approaches to installation? Which one is most expensive and risky?
8. A system costs RS 2,00,000 to install. The profit per year is RS. 50,000. Assuming an interest rate is 6%, what is the payback period of the investment?

Group C

Attempt any TWO questions[2x10=20]

9. List rules for diagramming DFDs. Draw a context diagram and expand it to top level logical DFD for Student Enrollment System. The enrollment process works as follows.
 - Students send in an application form containing their personal details and their desired course.
 - The university checks that the course is available and that the student has necessary academic qualifications.
 - If the course is available, the student is enrolled in the course, and the university confirms the enrollment by sending a confirmation letter that they are registered for the course.
 - If the course is unavailable, the student is sent a rejection letter.
10. Explain the factors that influence the cost of maintenance. Explain the steps involved in maintenance process and contrast them with the phases of system development lifecycle.
11. What qualities a software tester should have? What are the deliverables from coding, testing and Installation? Explain different types of Software testing in detail.

